



Programming Fundamentals

Class Assignment # 2

Total Marks: 60
Course Instructor: Faiza Khadim

Deadline: 23-04-2025: till 11:59 PM
Section: B8

Instructions:

- This Demonstrate knowledge of basic concepts, syntax of **user defined Functions and file handling** in programming (C++). So, you have to upload your assignments on CMS.
 - o CPP should be combined with files into zip folder.
- All program codes should be written in C/C++. Students should use Visual studio compiler for coding.
- **Don't use of any built-in function related to char array otherwise your task will be cancelled. Make your own functions for every single tasks.**
- **Don't use Global variables and string data type.**
- Plagiarism in the work is Professional dishonesty which leads to strict disciplinary action and zero in grading of Assignment.

CLO NO	CLO STATEMENT	PLO	Taxonomy LEVEL
2	Be able to apply fundamentals building blocks of a programming language to solve problems.	C3	2

Grading Rubric

Parameters	Weights	Remarks
Completeness	50%	Will determine by checking whether your code implements the tasks we ask to implement
Logic	10%	Logic building
Correctness	20%	Incorrect code behavior or not implementing something you are being told to.
Coding Standards	20%	Your code contains comments, well indented and easy to read. Make use of descriptive variable names and user prompts. For function-based programs the code should be organized in as many functions as required and each function contains single task.

Topic: File handling with functions, arrays

Section 1-10 Marks	
Objective	Able to recognize parameters and return types, as well as construct custom functions on their own that are frequently utilized.

Write the prototypes and the implementations of the following functions listed below:

- Sorting: according to Alphabetic character
- compare_Words: will compare two words for the exact match excluding the case. (Note lower and upper case both should mean the same.
- find_max: will find and returns the maximum value from the list of n values
- count_Character: will count all the characters in a CString.
- search: will find a character array value from the list of n values and returns true if it exists else return false.
- Strcpy(): function copies data from one C-style string into another C-string.

Section 2:[20+10+20=50 Marks]	
Objective	Capable of using the user-defined functions implemented above in their code without having to re-write the same code.

Question 1: Divide and Conquer [20-Points]

The PF teacher at your university needs help in grading a True/False quiz. The students' IDs and test answers are stored in a file. The first entry in the file **named "test.txt"** contains answers to the test in the form:

Answer Key: TFFTFFTTTTFFTFFTFTT

Every other entry in the file is the student ID, followed by a blank, followed by the student's responses. For example, the entry:

F22BSCS54301 TFTFTFTT TFTFTFFTTFT

indicates that the student ID is F22BSCS54301 and the answer to question 1 is True, the answer to question 2 is False, and so on. This student did not answer question 9. The exam has 20 questions, and the class has more than 150 students. Each correct answer is awarded two points, each wrong answer gets one point deducted, and no answer gets zero points. Write a program that processes the test data. The output should be stored in the file named **"result.txt"** such as the student's ID, followed by the answers, followed by the test score, followed by the test grade.

Assume the following grade scale:

90%–100%, A;
80%–89.99%, B;
70%–79.99%, C;
60%–69.99%, D;
0%–59.99%, F

Additional Task: This task should be Menu driven

- Display all records
- Find the roll number who has the lowest marks.
- Display the result according to marks/Grade wise (descending order).
- Search according to Roll number.

Sample Output:

```
F22BSCS54301 TFTFTFTTTFTFTFFTTT T 37.5 F
F22BSCS54302 TFFTFFTTTFTFTFTFTT 100 A
```

Menu:

1. Display all records
2. Find the student with the lowest marks
3. Display results sorted by marks/grade
4. Search by roll number
5. Exit

Enter your choice: 1

All Student Records:

```
F22BSCS54301 TFTFTFTTTFTFTFFTTT T 37.5 F
F22BSCS54302 TFFTFFTTTFTFTFTFTT 100 A
```

Enter your choice: 2

Student with Lowest Marks:

```
F22BSCS54301 TFTFTFTTTFTFTFFTTT T 37.5 F
```

Enter your choice: 3

Sorted Results (By Marks):

```
F22BSCS54302 TFFTFFTTTFTFTFTFTT 100 A
F22BSCS54301 TFTFTFTTTFTFTFFTTT T 37.5 F
```

Enter your choice: 4

Enter roll number: F22BSCS54302

Student Found:

ID: F22BSCS54302 Responses: TFFTFFTTTFTFTFTFTT Score: 100 Grade: A

Enter your choice: 4

Enter roll number: F22BSCS54310

Student Not Found

Enter your choice: 5

Exiting program.

Question 2: **Using divide and Conquer approach** **[10-Points]**

You have to create an application **spell checker** based on your programming fundamentals expertise.

You have to read data from file **input.txt** and check if all the words in file are spelled correctly.

Write a C++ program that finds all the misspelled words in a given text file **“input.txt”**. The misspelled words are found by checking against a file **“dictionary.txt”** which contains all possible correct words. Print all the misspelled words on the console screen.

Input.txt	Dictionary.txt
the name of my country is pakistan It is an Islamic coutry its nationel languge is urdu	Is It its an my the they name of at country city pakistan islamic national language urdu

Expected Output

Following words are misspelled in given file:

coutry
national
languge

Question 3: **Divide and Conquer** **[20-Points]**

Write a C++ program that reads a text file named “**sentence.txt**”, analyzes it, and writes a modified version to a new file named “**word_frequency.txt**”. The program should **count the occurrences of each word** in the original file and output this information to the new file, with each word and its count on a new line.

Sentence.txt

Hello world! This is a test file. This file is used
for file handling. File handling is interesting.
Hello again, world!

word_frequency.txt

Hello - 2
world - 2
This - 2
is - 3
a - 1
test - 1
file - 2
used - 1
for - 1
handling - 2
interesting - 1
again - 1
